

GeordieDaz AI

Integration Roadmap & Connected Services Plan

CONFIDENTIAL — CLIENT REVIEW

Prepared: 30 August 2026

Phase 1 — Core Voice AI COMPLETE

Feature	Status	Details
Real-time voice conversation	 Built	OpenAI Realtime API with WebSocket bridge
Alan Robson Geordie persona	 Built	Custom system prompts with thick dialect
Driving Mode persona	 Built	High-energy sarcastic co-pilot
Persistent memory (cross-session)	 Built	PostgreSQL + pgvector + Redis
JARVIS HUD interface	 Built	3-panel holographic UI with canvas voice orb
User authentication	 Built	JWT tokens with bcrypt hashing
ElevenLabs voice clone	 Built	Alan Robson voice clone created
Cloud deployment	 Live	Backend on Render, Frontend on Vercel

 **Currently blocked by:** OpenAI API credit balance (£0 remaining). Once credits are topped up, the full voice experience works end-to-end immediately.

Planned Integrations — Roadmap

Proton Mail

Capability	Example Voice Command
Read unread count	"You've got 3 new emails, pet."
Summarize emails	"One from your accountant about tax returns, two from Amazon."
Compose & send replies	"Send a reply saying 'Cheers, I'll look at it tomorrow.'"
Search inbox	"Find that email from Dave about the meeting."

How: Connects via Proton Mail Bridge (official IMAP/SMTP gateway). Emails parsed & summarized by LLM, responses spoken back.

Social Media — X (Twitter) & Instagram

Platform	Capability	Example
X / Twitter	Read mentions & DMs	"You've got 2 new mentions on X."
X / Twitter	Post tweets	"Post: Howay the lads! NUFC forever."
X / Twitter	Search trending	"What's trending in Newcastle?"
Instagram	Check DMs & notifications	"You've got 1 new DM on Instagram."
Instagram	Engagement stats	"Your last post got 150 likes."

How: Twitter API v2 & Instagram Graph API (OAuth 2.0). User authenticates once, tokens stored securely.

Crypto & Investment Tracking

Capability	Example Voice Command
Live crypto portfolio	"Your Bitcoin's at £42,150 — up 2.1% today, canny!"
Price alerts	"Bitcoin just hit £45,000, bonny lad!"
Stock tracking	"Your Tesla shares are up 3.2% this week."
Portfolio summary	"Overall portfolio is up 2.3% this month."

How: CoinGecko / CoinMarketCap for crypto, Alpha Vantage / Yahoo Finance for stocks. Prices cached in Redis, refreshed every 5 minutes.

 Future Integrations

Integration	Purpose	API
Google Calendar	"What's on me schedule today?"	Google Calendar API
Spotify	"Play some Geordie anthems"	Spotify Web API
Weather	"What's the weather in Newcastle?"	OpenWeatherMap
News	"What's the latest Toon news?"	NewsAPI / RSS
Smart Home	"Turn off the living room lights"	Home Assistant / Hue
WhatsApp	"Read my latest WhatsApp messages"	WhatsApp Business API

Plugin Architecture

Every integration follows the same modular pattern:

- 1. **Service file** — handles API communication
- 2. **Realtime tool** — GeordieDaz can call it mid-conversation by voice
- 3. **UI card** — shows live data in the Connected Services panel
- 4. **Memory integration** — remembers your preferences across sessions

Memory Persistence — Architecture

GeordieDaz uses a **3-tier memory system** that persists permanently across sessions, devices, and time:

Tier	Storage	What	Persistence
Working Memory	Redis	Last 20 conversation turns	24hr TTL, survives restarts
Semantic Memory	PostgreSQL + pgvector	Extracted facts (name, car, preferences)	Permanent — forever
Episodic Memory	PostgreSQL	Session summaries	Permanent — forever

Cross-Session Example

Session 1: User says "I drive a red Ferrari" → fact stored permanently in PostgreSQL with vector embedding

Session 2 (days later, different device): User says "What car do I drive?" → vector search finds the fact → GeordieDaz responds: "You drive a red Ferrari, bonny lad!"

Real-Time Memory Tools

Tool	Purpose	Example Trigger
search_memory	Search stored facts	"What car do I drive?"
store_fact	Store a new fact	"Remember I have a dog called Max"
search_history	Search past sessions	"What did we talk about last time?"

Handoff Confirmation

All of the following will be included in the final handoff:

Deliverable	Included
Complete source code (frontend + backend)	✓ Yes
All system prompts & persona configurations	✓ Yes
Memory extraction & retrieval logic	✓ Yes
LangGraph agent pipeline (7 nodes)	✓ Yes
Database schemas & migrations (Alembic)	✓ Yes
Docker Compose for local development	✓ Yes
Deployment configs (Render + Vercel)	✓ Yes
Environment variable templates	✓ Yes
API documentation (FastAPI /docs)	✓ Yes
Character Bible (Alan Robson guide)	✓ Yes
28 automated tests	✓ Yes
README with setup instructions	✓ Yes

